

Carmen AMO ALONSO

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RESEARCH INTERESTS

- Control Theory, Artificial Intelligence, Machine Learning, Convex Optimization.
- Large-Scale Systems, Data-driven Approaches, Foundation Models, Safety Guarantees.
- Natural Language Processing, Cyber-Physical Systems, Robotics.

PROFESSIONAL EXPERIENCE

ACADEMIC EXPERIENCE

Assistant Professor <i>Department of Electrical Engineering and Computer Sciences, UC Berkeley</i>	Incoming 2027 <i>Berkeley, CA</i>
Schmidt Science Fellow <i>Aeronautics & Astronautics Department, Stanford University</i>	2024 – 2026 <i>Stanford, CA</i>
Postdoctoral Fellow <i>Artificial Intelligence Center, ETH</i>	2023 – 2024 <i>Zürich, Switzerland</i>

INDUSTRY EXPERIENCE

Graduate Student Intern <i>Modeling and Optimization Engineering Team, Tesla Inc.</i>	2022 <i>Palo Alto, CA</i>
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EDUCATION

Ph.D. in Control and Dynamical Systems <i>California Institute of Technology</i>	2017 – 2023 <i>Pasadena, CA</i>
M.Sc. in Space Engineering <i>California Institute of Technology</i>	2016 – 2017 <i>Pasadena, CA</i>
B.Sc. in Aerospace Engineering <i>Technical University of Madrid</i>	2012 – 2016 <i>Madrid, Spain</i>

SELECTED HONORS

AWARDS AND RECOGNITIONS

MIT Brain and Cognitive Science Rising Stars Award - <i>Awarded to 3 postdocs per year worldwide</i>	2025
Best Paper Award of the IEEE Transactions on Control of Network Systems	2024
Milton and Francis Clauser Doctoral Prize - <i>For the best PhD thesis at Caltech across all disciplines</i>	2023
Best Student Paper Award at the International Conference on Control and Automation	2022
Named Rising Star in Electrical Engineering & Computer Science	2022
Named Rising Star in Cyber-Physical Systems	2022

FELLOWSHIPS AND SCHOLARSHIPS

Lucy Hsu Faculty Fellowship - <i>Awarded to a College of Engineering Assistant Professor, UC Berkeley</i>	2027 – 2032
Fellow of Stanford Impact Labs - <i>Awarded to 15 postdocs per year at Stanford</i>	2025
Emerson Consequential Scholarship (Emerson Collectives) - <i>Awarded to 1% of postdocs at Stanford</i>	2025
Schmidt Science Fellowship - <i>Awarded to 30 postdocs per year worldwide across all disciplines</i>	2024
MIT Engineering Excellence Postdoctoral Fellowship - <i>Awarded to 15 postdocs per year worldwide (declined)</i>	2023
ETH Artificial Intelligence Center Postdoctoral Fellowship - <i>Awarded to 12 postdocs per year worldwide</i>	2023
Amazon AI4Science Fellowship	2021
D.E. Shaw Exploration Fellowship	2019

MENTORING AND TEACHING EXPERIENCE

GRADUATE MENTORING

Stanford University	Palo Alto, California
<i>Mentor of master student Lukas Haverbeck</i>	2026
<i>Mentor of master student Ambroise Aigueperse</i>	2026
<i>Mentor of master student Enrico Vittori</i>	2026
ETH	Zürich, Switzerland
<i>Mentor of master student Federico Arangath – led to publication [C4] (7% acceptance)</i>	2024
<i>Mentor of master student Lukas Schuepp – led to publication [C3]</i>	2024
<i>Mentor of master student Alessandro Tuccillo</i>	2024
<i>Mentor of master student Gaia Lauper</i>	2024
<i>Mentor of master student Seif Ismail – led to publication [C7]</i>	2023
California Institute of Technology	Pasadena, CA
<i>Certificate of Interest in Research Mentoring</i>	2020

UNDERGRADUATE AND GRADUATE TEACHING

ETH	Zürich, Switzerland
<i>Co-instructor of Distributed Model Predictive Control</i>	Spring 2024
<i>Co-instructor of AI Center Projects in Machine Learning Research</i>	Winter 2024
California Institute of Technology	Pasadena, CA
<i>Teaching Assistant of Robust Control Theory (CDS 231)</i>	Spring 2023
<i>Teaching Assistant of Optimal Control and Estimation (CDS 112)</i>	Winter 2023
<i>Teaching Assistant of Network Control Systems (CDS 141)</i>	Spring 2021
<i>Teaching Assistant of Robust Control Theory (CDS 231)</i>	Winter 2020
<i>Head of Teaching Assistants of Introductory Methods of Applied Mathematics (ACM 95/100)</i>	Winter 2019
<i>Head of Teaching Assistants of Introduction to Probability Models (ACM 116)</i>	Fall 2019
<i>Teaching Assistant of Introductory Methods of Applied Mathematics (ACM 95/100)</i>	Winter 2018

PUBLIC ENGAGEMENT, OUTREACH AND SERVICE

PUBLIC ENGAGEMENT & LEADERSHIP

- AGI Governance Fellow**, Johns Hopkins University School of Government and Policy 2026
Selected in an intensive program in Washington, D.C. shaping the next generation of leaders in AGI governance.
- Engagement and Education Officer**, Stanford Science Policy Group, Stanford University 2025 – 2026
Leading educational and community outreach efforts connecting STEM research with policy-making.
- Member**, Future of Life Institute 2024 – Present
Part of a global community promoting the responsible development of transformative technologies.
- Fellow**, LEAP Alliance 2022 – 2023
Selected as part of a national initiative to diversify leadership in the computing professoriate.
- Co-founder**, Hispanics in Computing and Mathematical Sciences, California Institute of Technology 2021
Created an affinity group to support Hispanic students and lead outreach in underrepresented communities.

OUTREACH & EDUCATION

- Volunteer Instructor**, Clubes de Ciencia 2023 – Present
*Ongoing engagement with a nonprofit promoting science education for underrepresented students in Latin America.
Designed and taught a one-week course on Large Language Models at Merida Autonomous University (Mexico) and
National University of Engineering (Peru).*

ACADEMIC SERVICE & INTERDISCIPLINARY COMMUNICATION

- Co-organizer**, Workshop at the Conference on Robot Learning Austin, TX, 2026
The Science of Physical AI Safety
- Keynote Speaker**, NSF Workshop Washington D.C., 2025
NeuroAI and Beyond
- Co-organizer**, Workshop at IEEE Conference on Decision and Control Rio de Janeiro (Brazil), 2025
Large Language Models and Control
- Co-organizer**, Workshop at the International Conference on Learning Representations Vienna (Austria), 2024
Next Generation of Sequence Modeling
- Co-organizer**, Workshop at IEEE Conference on Decision and Control Cancun (Mexico), 2022
System Level Synthesis and its Frontiers

PUBLICATIONS

* denotes equal contribution

JOURNAL PUBLICATIONS

- [J1] E. Cheng and **C. Amo Alonso**. Linearly Controlled Language Generation with Performative Guarantees. *Transactions of Machine Learning Research (TMLR)*, 2026
- [J2] J. S. Li and **C. Amo Alonso**. Global performance guarantees for localized model predictive control. *IEEE Open Journal of Control Systems*, 2:325–336, 2023
- [J3] **C. Amo Alonso**, J. S. Li, J. Anderson, and N. Matni. Distributed and localized model predictive control. Part II: Theoretical guarantees. *IEEE Transactions on Control of Network Systems*, 10(3):1113 – 1123, 2023. **Best Paper Award**
- [J4] **C. Amo Alonso**, J. S. Li, J. Anderson, and N. Matni. Distributed and localized model predictive control. Part I: Synthesis and implementation. *IEEE Transactions on Control of Network Systems*, 10(2):1058 – 1068, 2023
- [J5] **C. Amo Alonso***, F. Yang*, and N. Matni. Data-driven distributed and localized model predictive control. *IEEE Open Journal of Control Systems*, 1:29–40, 2022

- [C1] E. Cheng, **C. Amo Alonso**, F. Danieli, A. Blaas, L. Zappella, P. Rodriguez, and X. Suau. A Formal Controllability Toolkit for Black-Box Generative Models. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2026
- [C2] A. Karuvally, F. Nowak, A. T. Keller, **C. Amo Alonso**, T. J. Sejnowski, and H. T. Siegelmann. Bridging Expressivity and Scalability with Adaptive Unitary SSMs. In *Proceedings of the Conference on Neural Information Processing Systems (NeurIPS)*, 2025
- [C3] L. Schüepp, G. De Pasquale, F. Dörfler, and **C. Amo Alonso**. System Level Synthesis for Affine Control Policies: Model Based and Data-Driven Settings. In *Proceedings of the 64th IEEE Conference on Decision and Control*. IEEE, 2025
- [C4] F. A. Joseph, J. Sieber, M. N. Zeilinger, and **C. Amo Alonso**. Lambda-Skip Connections: the architectural component that prevents Rank Collapse. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2025
- [C5] **C. Amo Alonso***, J. Sieber*, and M. N. Zeilinger. State Space Models as Foundation Models: A Control Theoretic Overview. In *Proceedings of the 2025 IEEE American Control Conference*. IEEE, 2025
- [C6] J. Sieber*, **C. Amo Alonso***, A. Didier, M. N. Zeilinger, and A. Orvieto. Understanding the differences in Foundation Models: Attention, State Space Models, and Recurrent Neural Networks. In *Proceedings of the Conference on Neural Information Processing Systems (NeurIPS)*, 2024
- [C7] S. Ismail*, A. Arbues*, R. Cotterell, R. Zurbrügg, and **C. Amo Alonso**. NARRATE: Versatile Language Architecture for Optimal Control in Robotics. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE/RSJ, 2024
- [C8] **C. Amo Alonso** and S.-H. Tseng. Effective GPU Parallelization of Distributed and Localized Model Predictive Control. In *Proceedings of the 17th IEEE International Conference on Control and Automation*. IEEE, 2022. **Best Student Paper Award**
- [C9] J. S. Li, **C. Amo Alonso**, and J. C. Doyle. Frontiers in Scalable Distributed Control: SLS, MPC, and beyond. In *Proceedings of the 2021 IEEE American Control Conference*. IEEE, 2021
- [C10] **C. Amo Alonso**, J. Anderson, and N. Matni. Explicit Distributed and Localized Model Predictive Control via System Level Synthesis. In *Proceedings of the 59th IEEE Conference on Decision and Control*. IEEE, 2020
- [C11] **C. Amo Alonso** and N. Matni. Distributed and Localized Model Predictive Control via System Level Synthesis. In *Proceedings of the 59th IEEE Conference on Decision and Control*. IEEE, 2020
- [C12] S.-H. Tseng, **C. Amo Alonso**, and S. J. Han. System Level Synthesis via Dynamic Programming. In *Proceedings of the 59th IEEE Conference on Decision and Control*. IEEE, 2020
- [C13] **C. Amo Alonso**, D. Ho, and J. M. Maestre. Distributed linear quadratic regulator robust to communication dropouts. In *Proceedings of the 21st World Congress of the International Federation of Automatic Control*. IFAC, 2020
- [C14] N. Olsman, **C. Amo Alonso**, and J. C. Doyle. Architecture and trade-offs in the heat shock response system. In *Proceedings of the 57th IEEE Conference on Decision and Control*, pages 1096–1103. IEEE, 2018